

Instruction manual



An ISO CERTIFIED COMPANY

Mars EdPal Instruments Pvt. Ltd.

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DEFLECTION OF BEAM APPARATUS

INSTRUCTION MANUAL



Mars EdPal Instruments Pvt. Ltd.
AMBALA, (HARYANA)

APPARATUS REQUIRED:

- Simple Supported Beam Apparatus
- Dial Gauge : 1 No.
- Magnetic Base : 1 No.
- Hangers : 2 Nos. (with assembly □)
- Beams : 3 Nos.
- 100 gm weights : 5 Nos.
- 200 gm weights : 5 Nos.
- 500 gm weights : 10 Nos.
- 1 Kg weights : 5 Nos.
- Beams : 2 steel + 1 wooden
- Fixed assembly : 2
- Pillar : 1



Deflection Of Beams Apparatus

1.0 OBJECTIVE:

To determine the deflection of a simply supported beam when it is centrally loaded.

2.0 EQUIPMENT:

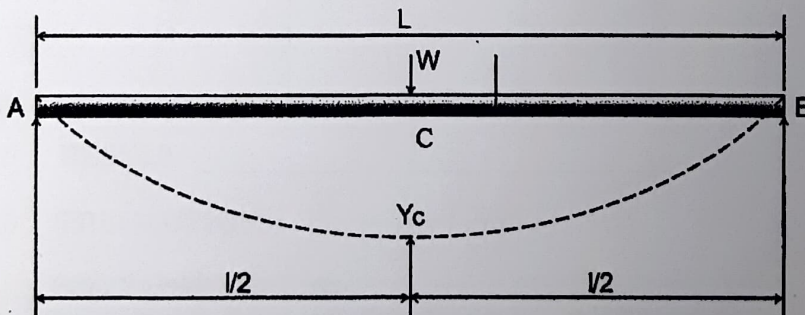
Apparatus consists of a M.S. beam, which is supported on two supports. Apparatus is centrally loaded. Actual deflection can be measured by the help of a dial gauge at the point of loading. Apparatus is supplied complete with weights and supporting stand.

3.0 INTRODUCTION AND THEORY:

Whenever a beam is loaded, it deflects from its original position. The amount, by which a beam deflects, depends upon its cross section and the bending moment. In modern design offices, there are two design criteria for a beam (a) strength and (b) stiffness.

As per the strength criterion of the beam design, it should be strong enough to resist bending moment and shear force. But as per the stiffness criterion of the beam design, it should be stiff enough to resist the deflection of the beam.

Consider a simply supported beam AB of length l and carrying a point load W at the center of beam C.



$$\text{Or } y_c = -\frac{WL^3}{48EI} \quad (\text{minus sign means that the deflection is downwards})$$

$$y_c = \frac{WL^3}{48EI} \text{ (Theoretical Deflection)}$$

4.0 EXPERIMENTAL PROCEDURE:

- Step1: Measure the distance L between the two supports.
- Step2: Note the initial readings of graduated scale.
- Step3: Put load W on the center of the beam. Now find R_A & R_B .
- Step4: Note readings of deflection y from the dial gauge.
- Step5: Calculate $y_{\text{calculated}}$ from the formulae.
- Step6: Deflection should be same as obtained analytically and experimentally.
Calculate % error.
- Step7: Repeat the experiment by taking different values of W.

5.0 OBSERVATIONS AND COMPUTATION SHEET:

- Material of the beam is = M.S, Aluminum & Wood
- Length of beam, L cm = 1120 & 1000 (wood) in mm
- Thickness of beam, t cm = 5 & 6 (wood) in mm
- Breadth of the beam, b, cm = 25 & 27 (wood) in mm
- Moment of Inertia of beam, I = $\frac{bt^3}{12} =$
- Modulus of elasticity of beam material, E = $2.0 \times 10^5 \text{ (kg/mm}^2\text{) (M.S) } \text{N/mm}^2.$
7036.04 kg/mm² (Alum)
1274.64 kg/mm² (Wood)

Aluminum

Sr. No.	Weight on beam W	Deflection observed y	Deflection Calculated y_c	% error
01				

6.0 RESULT: _____

7.0 PRECAUTIONS:

- Apply the loads without jerk.
- Perform the experiment away from vibration and other disturbances.
- Dial gage fixed at the bottom such that pointer touches base of weights hanger
- Remove all loads after conducting the experiment
- Initially apply 500gm wt and take its reading as reference before proceeding to apply actual weights

DEFLECTION OF BEAM APPARATUS

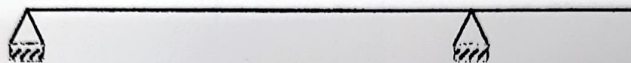
- No vibrations on the table

RECIPROCATING, CONTINUOUS & CANTILEVER BEAM APPARATUS

1.0 THEORY:

Continuous beam apparatus in its simplest form states that the deflections of any point A of any elastic structure due to a load P at any other point B is the same as the deflection of B due to the same load P applied at point A.

It is, therefore easily derived that the deflection curve for a point in a structure is the same as the deflected curve of the structure when unit load is applied at the point for which the influence curve was obtained.



2.0 OBJECTIVE:

To verify Clerk Maxwell's theorem by means of a mild steel beam.

3.0 APPARATUS:

Apparatus consists of a M.S. beam with graduations at every 10cm along the length. A dial gauge and a magnetic base are provided with the apparatus.

4.0 SUGGESTED EXPERIMENTAL WORK:

- Step 1: Apply a load either at the centre of the simply supported span or at the free end of the beam, the deflected form can be obtained.
- Step 2: Measure the height of the beam at certain distance by means of a dial gauge before and after loading and determine the deflection before and after at each point separately.
- Step 3: Now move a load along the beam at certain distance and for each positions of the load, the deflection of the point was noted where the load was applied in step 1. This deflection should be measured at each such point before and after the loading, separately.
- Step 4: Plot the graph between deflection as ordinate and position of point on abssica the plot for graph drawn in step 2 and 3. These are the influence line ordinates for deflection of the beam.

5.0 RESULTS AND DISCUSSIONS:

Compare the two curves, hence show the influence line curve for the deflection of the beam.

6.0 SAMPLE DATA SHEET:

Distance from the pinned end <i>4-4</i>	Load at central point <i>Point A</i> cantilever end		Deflection of various points (mm) 2-3	Load moving along the beam <i>Point B</i>		Deflection of various points (mm) 5-6
	Beam unloaded Dial gauge reading (mm) 2	Beam loaded Dial gauge reading (mm) 3		Beam unloaded Dial gauge reading (mm) 5	Beam loaded dial gauge reading (mm) 6	

7.0 COMMENT:

Weight on Beam	Deflection at B	Deflection at A
at Point A	 	
at Point B	 	

